

DSA STUDY BLUEPRINT SERIES

68. Next Greater Element II Stack Queue

Monotonic Stack Search on Circular Arrays

Section 08 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Problem:** Next greater element in a circular array.
- Simulate circular traversal by executing loops up to $2 \times N - 1$ steps.

Memory & Execution Blueprint

Circular array: [1, 2, 1] $\Rightarrow$ Next greater of last 1 wraps around to find 2.

C++ Implementation Reference

Standard code layout and syntax

```
vector nextGreaterElements(vector& nums) {  
    int n = nums.size();  
    vector ans(n, -1); stack st;  
    for (int i = 0; i < 2 * n; i++) {  
        while (!st.empty() && nums[i % n] > nums[st.top()]) {  
            ans[st.top()] = nums[i % n]; st.pop();  
        }  
        if (i < n) st.push(i);  
    }  
    return ans;  
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Time Complexity	$O(N)$
Space Complexity	$O(N)$

Key Practices & Gotchas

- **Important:** Using modulo operations (`i % N`) maps circular array indices correctly.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace monotonic stack modifications for next greater element on `[4, 5, 2]` :

Active Element	Stack State (Indices)	Pop comparison	Result output
4	[0] (value 4)	N/A	Waiting for next greater
5	[1] (value 5)	5 > 4 (pop index 0)	Next Greater of 4 is 5

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Monotonic properties:** Stacks process elements strictly increasing or decreasing, optimizing nested lookups.
- **Min Stack details:** Track current minimum bounds by storing elements as encoded difference values.
- **Related LeetCode Problems:** #155 Min Stack, #84 Largest Rectangle in Histogram, #42 Trapping Rain Water.